

Shubhangi Jeve

B.Tech in Artificial Intelligence & Data Science
Terna College of Engineering, Dharashiv | CGPA: 8.5
Hyderabad, Telangana

+91-9579122372
shubhangijeve@gmail.com

shubhangijeve.github.io/My-Portfolio
github.com/ShubhangiJeve
linkedin.com/in/shubhangi-jeve-97445b235






CAREER OBJECTIVE

AI Systems Engineer specializing in Agentic AI, Advanced RAG, and production LLM orchestration. Proven track record building end-to-end multi-agent architectures, implementing observability and automated evals (RAGAS, DeepEval, Langfuse), debugging failure modes, and optimizing inference cost and latency in enterprise environments.

EXPERIENCE

- AI Engineer Intern
COGNITBOTZ

November 2025 – Present
Hyderabad, Telangana

Project 1: LegalAID — Enterprise-grade AI Legal Research and Litigation Platform combining Hybrid RAG, semantic search, and automated drafting.

- Advanced Hybrid RAG & Reranking: Engineered dense-lexical retrieval (pgvector BGE-M3 + Postgres FTS) with FlashRank cross-encoder reranking; eliminated keyword retrieval misses and lifted Recall@5 from 72% to 94%.
 - Evals & Guardrail Hardening: Benchmarked generation across 500+ legal queries using RAGAS (faithfulness, context relevance); tuned abstention thresholds to catch retrieval hallucinations, slashing hallucination rates to <1%.
 - High-Throughput Ingestion: Automated ETL pipeline parsing 75,000+ court JSON cases over 25 years with parent-child hierarchical chunking, schema normalization, and deduplication.
 - Async LLM Gateway (FastAPI): Architected multi-model gateway (OpenRouter + NVIDIA NIM) with semantic prompt-caching and Pydantic v2 validation, cutting median latency by 45% and API costs by 38%.
 - Production Containerization: Containerized full stack via Docker Compose with health-check dependency chains, ensuring 99.9% uptime and data persistence.

Project 2: MeetOps — Enterprise AI Meeting Copilot and Project Intelligence Platform integrating Teams via ephemeral bots and agentic workflows.

- Transcript RAG & Speaker Chunking: Architected semantic RAG with speaker-turn chunking and pgvector embeddings, lifting topical retrieval precision by 35%.
 - Agentic Workflows & Tool Reliability: Engineered multi-step tool-calling agents (LangGraph/OpenAI) for Jira and SAP GRC; implemented deterministic fallback schemas and loop guards, boosting task success from 68% to 92%.
 - AI Observability & Tracing: Integrated distributed tracing to monitor tool latency, token burn, and step failures, diagnosing context fragmentation during 60+ min meetings.
 - Confidence Abstention Gating: Enforced 0.75 cosine similarity guardrail instructing LLMs to abstain on low context, eliminating fabricated technical details.
 - Low-Latency Streaming: Deployed multi-tier Redis semantic cache with SSE streaming, reducing perceived generation latency by 60% during live sessions.

Project 3: Granite Buyer Intelligence — Local-first B2B intelligence and lead generation platform automating buyer discovery, evidence qualification, and M365 outreach.

- Modular Monolith Architecture: Designed 9-domain lifecycle system (Discovery, Evidence, Intelligence, Email, Jobs) delivered solo from design to production.
 - Multi-Tier LLM Failover Gateway: Implemented 3-tier failover (OpenRouter Nemotron-550B → NVIDIA Build → Groq) and TF lexical search, guaranteeing 100% uptime.
 - Durable Task Queue: Built crash-resilient PostgreSQL job queue with 180s leases and heartbeat recovery, handling automated multi-engine web discovery.
 - Evidence-First Validation: Enforced SQL check constraints separating verified facts from AI inference, with 0–100 quality scoring and audit tracking.
- Artificial Intelligence Intern
Infosys Springboard

February 2025 – April 2025
Remote

Built a Healthcare RAG Chatbot on a Wikipedia dataset using FAISS vector search, Flask backend, and React/Tailwind frontend for grounded clinical Q&A.

Implemented a real-time query processing pipeline with source verification and an interactive UI supporting multi-turn question answering sessions.
- Data Science Intern
Adhyayan IT

January 2025 – June 2025
Remote

Built a Text-to-SQL system using LangChain and LLMs (Gemini, Groq) with RAGAS evaluation achieving 100% context precision and high helpfulness scores.

Developed an interactive Streamlit UI with secure MySQL connectivity, enabling non-technical users to query databases efficiently.
- Data Analyst Intern
Cyber Police Station

October 2023 – January 2024
Dharashiv, Maharashtra

Managed 500+ user credentials for the National Cyber Crime Portal and organized 1,000+ complaints, improving case resolution efficiency by 30%.

- Communicated with victims to provide timely status updates, enhancing overall satisfaction by 15%.

PROJECTS

Project 1: Argus — Multi-Agent AI Platform

Autonomous hierarchical multi-agent platform with specialized tool routing, runtime guardrails, and full observability

- **Hierarchical Agent Routing (LangGraph):** Architected supervisor-specialist framework decomposing goals across research, analysis, and writing agents with scoped tool permissions.
- **Observability, Tracing & Failure Debugging:** Instrumented full-lifecycle Langfuse tracing across every agent hop; diagnosed tool loops and context overflow, reducing step failure rate by 34%.
- **Agent Evaluation Test Suite:** Benchmarked multi-step planning accuracy and tool-selection precision against golden trajectories, improving task completion from 71% to 89%.
- **Dual-Ended Guardrail Pipeline:** Enforced input prompt-injection detection, strict tool allowlists, and Pydantic output schema validation with PII scrubbing.
- **Decoupled Async Backend & Deployment:** Built async FastAPI service with dynamic provider failover; deployed on Azure VM with Docker, Nginx, and TLS.

Project 2: LedgerLens-AI — Multi-LLM Document Intelligence & Forensic Engine

Production-grade multi-model orchestration engine leveraging Databricks Unity Catalog, FastAPI, and OpenRouter for high-throughput financial extraction and contract reconciliation

- **Multi-Model LLM Routing:** Engineered dynamic routing via OpenRouter directing forensic deduction to DeepSeek R1 and entity extraction to GPT-4o, cutting API inference costs by 42%.
- **Databricks Lakehouse & Unity Catalog:** Architected data pipelines on Databricks Unity Catalog and Delta Lake, enforcing data lineage, audit logging, and RBAC governance across 100,000+ financial documents.
- **Structured Output Validation & Caching:** Developed Pydantic schema-repair pipelines with PostgreSQL prompt-caching, achieving 99.4% adherence and reducing TTFT by 350ms.
- **Chain-of-Thought Forensics:** Implemented automated reasoning cross-examining balance sheets against vendor agreements, identifying \$1.8M+ in unapplied volume rebates.

Project 3: Domain-Adapted LLM Fine-Tuning & Evaluation Engine

Parameter-efficient post-training pipeline with QLoRA, automated G-Eval benchmarking, and optimized vLLM serving

- **QLoRA Fine-Tuning:** Fine-tuned open-weights LLMs (Llama-3-8B / Mistral) using Unsloth and Hugging Face TRL on domain instruction datasets, reducing structured output formatting errors by 82%.
- **Automated Evals & Benchmarking:** Evaluated checkpoints against baseline models using DeepEval/G-Eval (reasoning, syntax adherence), validating comparable accuracy to GPT-4o-mini at 75% lower inference cost.
- **Production Inference Optimization:** Quantized weights to 4-bit AWQ and served via vLLM with continuous batching and PagedAttention, achieving 3.8x higher throughput.

Project 4: Insurance Claim Automation System

AI-powered insurance claim processing using Google Gemini AI, LangChain, and Flask

- **Multimodal Claim Extraction:** Developed automated medical claim validation pipeline using Google Gemini Vision and NLP, reducing manual review time by 70%.
- **Decision Engine & Validation:** Built Flask web application with LangChain integration for intelligent claim assessment, Pydantic error handling, and confidence-based routing.

TECHNICAL SKILLS

Agentic AI & LLMs: Multi-Agent Systems, LangGraph, LangChain, Tool/Function-Calling, State Graphs, Supervisor-Worker Patterns, Prompt Engineering, Prompt Caching, OpenAI, Claude, DeepSeek R1, Gemini

Advanced RAG & Search: Hybrid Search (Dense + BM25/FTS), Cross-Encoder Reranking (FlashRank, BGE-Reranker), Parent-Child Chunking, pgvector (PostgreSQL), Weaviate, FAISS, ChromaDB

AI Evals & Observability: LLM Evals (RAGAS, DeepEval, G-Eval), Observability & Tracing (Langfuse, OpenTelemetry), Failure Mode Analysis, Guardrails (NeMo Guardrails, Pydantic, Abstention Gating)

Fine-Tuning & Inference: PEFT, LoRA/QLoRA, Unsloth, Hugging Face (Transformers, TRL), Quantization (AWQ/GGUF), vLLM, NVIDIA NIM, Model Routing & Failover

Backend, Cloud & MLOps: Python, FastAPI, AsyncIO, REST APIs, PostgreSQL, Docker, Azure, Databricks (Unity Catalog, Delta Lake, MLflow), CI/CD, Git

EDUCATION

● Bachelor of Technology in Artificial Intelligence and Data Science

Terna Public Charitable Trust's College of Engineering, Dharashiv

2021 – 2025

CGPA: 8.5